

Percent Change From Initial Price to New Price

s = scale factor

i = initial (old) price

n = new price

c = percent change (decimal form)

$$s = \frac{n}{i}$$

$$i = \frac{n}{s}$$

$$n = si$$

$$c = |s - 1|$$

- If new price is **more** than initial price, the percent change is a percent **increase**, and $s = c + 1$, and $c = s - 1$, and:

$$c = \frac{n}{i} - 1$$

$$i = \frac{n}{1 + c}$$

$$n = i \cdot (1 + c)$$

- If new price is **less** than initial price, the percent change is a percent **decrease**, and $s = 1 - c$, and $c = 1 - s$, and:

$$c = 1 - \frac{n}{i}$$

$$i = \frac{n}{1 - c}$$

$$n = i \cdot (1 - c)$$